

USER GUIDE

for

Cost Optimisation Model for Industrial Technologies (COMIT)

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1 Introduction

COMIT is an optimisation model of UK industry that finds least cost industrial decarbonisation pathways under user-defined constraints / assumptions.

The aim of this guide is to provide a short introduction to how to run COMIT, including how to set up and run the model, and download and interpret the outputs. Guidance is also provided on how to change the model input parameters and run the model under different types of user scenarios.

Please note that due to limited time and resources we are only able to provide a brief introduction to using COMIT. Unfortunately, we cannot respond to additional queries about the model.

2 Model overview

The following model map shows a flow of how the model works, with inputs read from an Excel workbook into an Rshiny app, where the objective function and constraints are set up, and the problem is solved as a linear programming model. The outputs are presented in an Excel workbook with information about the model's calculated (optimal) solution.



Figure 1 Flow of the COMIT model, from inputs to outputs.

2.1 Getting started

2.1.1 R package installation

COMIT can be installed as an R package from a local source file (e.g. **COMIT.1.0.0.tar.gz**). Full instructions on how to do this are provided in the **install_code.R** file.

If updates have been made to the COMIT package and a new release is available to install, instructions for how to do this are provided in the **update_code.R** file.

2.2. Running the model

1. Start the Rshiny app by installing the COMIT package and using the **library(comit)** and **run_app()** commands in R/Rstudio.
2. Obtain a copy of the Excel input workbook (e.g. from the Git repository).
3. Set up the desired scenario in the Excel Input workbook to be run with COMIT. This requires making any necessary changes to the default parameter values and data contained in the provided copy of the Excel input data workbook (how to do this is described in more detail in section 5).
4. Once the input data workbook is saved and ready, the model can be run by clicking on the browse button and selecting the prepared workbook file to upload to the app.

5. Once the file is uploaded, the model will message “*Uploading file*” in the app window and begin to run. The progress bar will indicate progress through the model run and the app will message progress updates in the app window (“*Reading spreadsheet*”, “*Creating decision variables*”,... etc.) as it completes each stage of the model process. The model should take about 5 minutes to run at a 5-year timestep and 30-60 minutes to run with a 1-year timestep¹.

Once completed, the app will sound a tone to indicate it has finished running and download a .zip file containing the results in an Excel workbook. The Excel output file will have the same name as the Excel input file, but with the prefix ‘Scenario_’ added to allow for easy identification (e.g. Scenario_comit_input_scenario1a.xlsx).

2.2. Running multiple scenarios at once

Multiple input files can be uploaded at once. To run the model with multiple input files:

1. Set up and save a separate Excel input file for each of the scenarios you wish to run. For example, if you had three different scenarios you wish to run, you would place three separate input workbooks (e.g. “Data_template_complete_scenario1a.xlsx”, “Data_template_complete_scenario1b.xlsx”, etc.) in the same folder ready to be uploaded.
2. Next, start the COMIT app and click the ‘Browse’ button. Navigate to the folder where the input files are saved, hold down the shift key to select multiple files at once, and then click the Open button to upload them all to the app.
3. The app progress bar will indicate progress through the inputted files, and once all scenario runs are completed, the model will allow the user to download a single .zip file containing separate output files for each of the inputted scenario files.

Note that the ‘multiple runs’ feature has so far only been tested with 5 scenarios at once. Uploading too many scenarios at once is not recommended as it may cause the model to run out of memory and fail to complete all runs.

2.2. Viewing outputs in the app

Once the COMIT app has finished all scenario run(s) and the .zip output file has been downloaded, the user then has the option to view summary graphs for key outputs in the app window. To do this, click on the ‘Outputs’ tab in the bar at the top of the COMIT app window (the ‘Model’ tab is the default view when the app is started and is the one that enables users to upload Input files and run the model)

If multiple scenarios were uploaded, there will be the option to view and compare them. Similarly, if the counterfactual run parameter is set to TRUE, the option to display the corresponding counterfactual scenario will also be provided. In addition, the user has the option to upload additional output files from other runs, to add to the scenarios being compared.

There is also the option to directly download a spreadsheet containing the data being used to generate the graphs in the app window (with this feature, the user is downloading a subset of the full set of results in the Output workbook for each scenario selected)

¹ However, note that run time can increase considerably for 1-yearly runs where very large numbers of user constraints have been added, as this increases the complexity of the LP being set up and solved.

3 The input workbook

3.1 Overview

COMIT uses a single Excel workbook to contain all the input data and assumptions. This input workbook allows users to make changes to model controls, constraints, parameters and input assumptions in order to run the model for different user-specified scenarios.

An overview and brief description of the contents of the Excel Input workbook is provided in this section of the user guide. Guidance on how to change the input workbook to run different user scenarios is provided in section 5. *Developing and testing scenarios*.

The input workbook is made up of 4 different sections of worksheets: (i) Parameters, (ii) Constraints, (iii) Assumptions, and (iv) Data (

Table 1).

Table 1 An overview of the 4 sections in the Excel Input workbook.

Section	Name	Contents
1	Parameters	The tabs in this section contain key parameter settings that are used to specify the model.
2	Constraints	The tabs in this section contain model controls and user constraints.
3	Assumptions	The tabs in this section contain technology and data assumptions.
4	Data	The tabs in this section tidy up site level data, NAEI sites and non-point sources (non-traded sites) based on ONS payee data and IDBR data.

Important tabs to be aware of are the model parameters tabs (**model_parameters_a**, **model_parameters_b** and **model_parameters_c**) and the **constraints_to_include** tab.

The **model_parameters** tabs allows the user to specify various model run options, such as the model timestep length, and the start and end year to use for the model run. The **constraints_to_include** provides a list of all the constraints in the model and allows the user to switch them either on or off. If a constraint's 'include' value is set to FALSE, the model will ignore any values in the relevant sheet, which means that when setting up or editing a new constraint, the user should always first make sure it is switched on in this tab.

The **Technologies** and **technology_input_output** tabs in the **Assumptions** section list the industrial technologies included in the model. A brief overview of the naming conventions used for the technologies and industrial commodities included in the model is provided in **Appendix A**. These can help users with interpreting both the inputs and outputs from COMIT.

4 Understanding COMIT Outputs

The model outputs a least cost optimal pathway for the industry sector including refineries (and hydrogen production when required) between 2021–2050. As part of this optimal solution, it calculates technology deployment pathways for each industrial site in 1 or 5 yearly intervals along with the associated infrastructure and transportation requirement (e.g., CO2 transport pipes) and costs.

COMIT summarises these optimal pathways into a single Excel output file, providing outputs for the associated Costs, Used industrial capacity, Emissions, and Energy use at both industrial cluster level and individual industrial site level where applicable/desired.

The COMIT Excel output file contains the following worksheets:

1. **Title** [dark blue]: Workbook title page

2. **Info** [dark blue]: a high-level overview of the contents of each of the sheets (tabs) in the output workbook.
3. **Definitions** [dark blue]: definitions for each of the field headings in the raw data output sheets.
4. **DetailedTables** [purple]: this sheet contains the key summary results tables, compiled from the raw output data in the 6 light blue tabs.
5. **Sites_info** [light blue]: this sheet contains data for the individual sites that are modelled, including some high-level characteristics and geographical information
6. Raw data outputs [light blue]: contains the site-level outputs for **Costs**, **Emissions**, **Outputs**, **Energy**, and **Infrastructure**.
7. Cluster-level outputs [orange]: if the option to output cluster-level data is selected [see **model_parameters_b** sheet in the Input workbook], the output file will also contain 5 additional sheets with the corresponding site-level outputs [**Costs_cluster**, **Emissions_cluster**, **Outputs_cluster**, **Energy_cluster** and **Infrastructure_cluster**]
8. **Log**: this tab contains a record of any warnings or error messages recorded whilst COMIT runs.

4.1 Info tab

This provides a high-level overview of the contents of each of the sheets (tabs) in the output workbook

4.2 Definitions sheets

The **Definitions** sheet provides accompanying descriptions for each of the field headings and categories used in the cluster level data output sheets.

4.3 Detailed Tables

This sheet provides tables of outputs through time summarised by sector and other headline categories, such as emissions source/type, technology, and types of capital costs.

For all tables in the **DetailedTables** sheet, the individual data points that are used to generate these summary tables can be accessed from the corresponding blue tab.

For the Emissions table there is the option in cell B21 to select whether to view direct and indirect emissions or just direct emissions (which exclude emissions from electricity and hydrogen production).

Note that if a 5-year timestep is used, the output file will only show datapoints at 5-yearly intervals, and other years will be left blank.

4.4 Sites_info

This sheet contains data for the individual sites that are modelled, including some high-level characteristics and geographical information.

Sites in COMIT will either be large point source sites taken directly from the NAEI database, or an artificial site created by COMIT for modelling purposes. NAEI sites will have PlantIDs that can be used to directly link them back to the NAEI input data in the Excel input workbook (the **NAEI_df_clean_2021_revised** tab). Sites without a PlantID are artificial sites.

4.5 Costs

This sheet displays all site-level outputs relating to costs incurred during the model run. Units are in £million. All costs presented in the output spreadsheet are undiscounted (but they are discounted when they are processed through the model).

There are two main types of costs being displayed: (i) the costs associated with building and using industrial *Technologies* to generate output commodities, and (ii) the *Infrastructure* costs of building and using pipes and/or trucking or gridlines to transport H2 and CO2 between sites, clusters and storage sites. The **Sector_infrastructure** field will list the corresponding industrial sector if the cost was incurred from a technological process, or the type of infrastructure if it is a cost associated with pipes, trucking or gridlines (note the following codes used for cluster to cluster/terminal transport, **CO2_C2S**: CO2 cluster to storage costs, **H2_C2C**: H2 cluster to cluster costs).

The following different types of costs are included (**Cost_type** column): Capital expenditure cost with borrowing and spread over technology/equipment lifetime (capex), Operational costs (opex), Fuel Cost, Carbon Cost and 'lumpy' yearly investment cost figures without borrowing (capex_lump).

The information displayed in the **DetailedTables** summary tab is calculated by summing these individual cluster level outputs by, for example, Cost_type and Sector_group (Capex_lump is excluded to avoid double counting of capex costs).

4.6 Emissions

This tab shows Greenhouse gas emissions from technologies in Kt of CO2 equivalent through time (KtCO2e). As with the Costs tab, the field headings in the top row can be used to select data for particular groupings of data, including: groupings for the technology which releases the emissions and groupings for the emission itself (such as traded / non-traded).

4.7 Outputs

This tab shows the deployment of industrial technologies in terms of used capacity and the activity through time. This sheet is useful to understand the roll out of technologies. As with the Costs tab, the field headings in the top row can be used to select data for particular groupings of outputs.

4.8 Energy

This tab shows the amounts of fuel used at the breakdown level (by technology / cluster / site etc.) along with the corresponding fuel emissions. Fuel definitions can be found on the **DetailedTables** sheet (from row 118). This sheet contains the same information as the Emissions sheet but with the type of pollutant (CO2, methane, etc.) included. As with the Costs tab, the field headings in the top row can be used to select data for particular groupings of data.

Fuel use is reported for each year in both Terawatt-hours (TWh) [column headings 2021_TWh, 2022_TWh,...] and Kt of CO2 equivalent through time (KtCO2e) [column headings 2021_ktCO2e, 2022_ktCO2e,...].

If you are interested in calculating total energy use (for example, by sector), you will need to remove the **NonEnergyUse** Fuel use category.

4.9 Infrastructure

This tab shows the capacities of H2 and CO2 pipes built and the amount and transport type for H2 and CO2 transported between sites, clusters and storage sites. As with the Costs tab, the filters in the top row can be used to select data for particular groupings.

4.10 Cluster-level outputs

If the user selected to output cluster level data in the **model_parameters_b** tab of the Excel input workbook, there will also be cluster-level versions of the above site-level worksheets.

5. Developing and testing scenarios

The following section outlines some practical examples of how to modify the input parameters contained in the input workbook to use COMIT to explore various scenarios and sensitivities surrounding industry decarbonisation.

If you are interested in modifying a particular tab in the input workbook – searching for the tab name in this guide (e.g. ‘**supply_chain_constraints**’) will be a quick way to identify the section that addresses how to make changes to information in that particular tab.

For each scenario below, the scenario description will illustrate how to change a subset of the parameters in the input workbook. All parameters not explicitly mentioned in the scenario vignette should be assumed to remain at their default values, as specified in the Excel input workbook.

While COMIT is quite flexible in being able to represent different scenarios, the user will still always need to be careful about the overall coherence of the various assumptions underlying a particular scenario, otherwise the model may not be able to find a solution.

General tips for developing and testing new scenarios:

1. Inputs need to be sensible. Making changes to the input parameters may create situations where the model is no longer able to solve for an optimal solution. Users must be careful not to over constrain the model. For example, setting the overall industrial emissions constraint to an unrealistically low level will not allow the model to solve. This means that care is necessary in developing *coherent* scenarios, and that some trial and error may be necessary to arrive at a scenario that can be solved.
2. To test the impact of new scenarios it can be helpful to change inputs or constraints gradually, (e.g., one at a time) rather than all at once. This makes it easier to understand the impact of each change on the model and avoids unnecessary confusion when interpreting the outputs.
3. It can be valuable to first test out a scenario using the 5-year timestep interval. This allows for quick feedback on whether the model can solve under a particular scenario. Once this is confirmed, the more time intensive 1-year timestep version can then be run. In some cases, using 2-year timesteps can also be useful, as it provides greater granularity, but solves in a much faster time than a 1-yearly scenario.

Scenario A: Running the model with a different timestep

The basic time unit in the model run is the *timestep*. In COMIT the length of a timestep is defined in years. A five-year timestep is the default choice specified in the Input data file in the **model_parameters_a** tab.

There are two main reasons for choosing a >1-year timestep: (i) shortening the processing time for model runs that are otherwise unreasonably long, and (ii) allowing the use of parameters or input data that have been estimated for multi-year periods and cannot easily be rescaled. The trade-off for selecting a larger timestep is that the outputs will be more granular: the outputs in the output file will reflect the choice of timestep (e.g., if a 5-year timestep is chosen, then the model outputs will be in 5-year increments).

Running the model with a timestep of 1 (year) to allow for yearly outputs is the scenario that users are mostly likely to be interested in for this parameter. The model can also run with a 2-year timestep, which can be useful by providing a reasonably fast model run with greater granularity than a 5-year timestep run.

For most tabs, the default values in the input workbook contain data ready for running the model with 1-year timesteps (e.g. 2021, 2022, 2023,...,2051). When a 5-year timestep is selected, the model then runs using just the data at 5-yearly intervals (e.g. 2021, 2026, 2031, etc...) and ignores the other timestep values. However, the model can also interpolate data for missing years using linear interpolation, and for some tabs for which there is only 5-yearly input data (e.g. the

minimum_capacity tab), this is what will happen when a yearly-run is carried out using the default Input file. Importantly the model can handle input data in 1-year increments for some tabs and 5-year increments for others for both 1-year and 5-year timestep runs.

Scenario B: Constraining technology deployment for a sector/output

The user may be interested in simulating a scenario where the deployment of a particular technology is restricted. For example, to represent situations where a technology is not likely to be available, or not available at scale/across all sectors or regions represented in the model during certain years of the COMIT run period.

The **maximum_capacity** tab can be used for this purpose, to allow the user to limit the maximum proportion a technology can contribute to the production of its respective output in each year

For example, it may be desirable to restrict hydrogen deployment during the 2020s and 2030s (when most H2 fuel switching is expected to take place in Industry's pathway to Net Zero) to mimic the effects of supply chain constraints limiting the speed with which hydrogen technologies can be rolled out across industrial sites.

To simulate a scenario where H2 deployment is constricted, a constraint can be added to the **maximum_capacity** tab for all hydrogen technologies.

To set up this scenario (or a similar one) in the input workbook:

1. From the **Technologies** tab identify the relevant technology codes. In this instance, all technologies for which the **technology_category** field is "Hydrogen" are selected.
2. In the **maximum_capacity** tab, input each technology code into its own column (in row 7) starting from the first blank column and populate the remaining cells in each column with the maximum percentage of output capacity for each year from 2021 to 2050.

Scenario C: Changing carbon values

By default, the model runs with a carbon cost incorporated into the model. The user can change the values in this tab to explore alternative carbon values and their impact on the model's outputs.

Scenario D: Running the model with no carbon value and an emissions constraint

By default, COMIT runs with a carbon cost included in the objective function. An alternative approach to using COMIT is to run the model without a carbon value included in the objective function and instead specify specific emissions constraints that must be met for each year of the model run. In this mode, the model is then finding the least cost solution that satisfies a desired level of emissions reduction (rather than using a carbon value to drive emissions reductions, as in the standard way of using the model).

The carbon cost can be easily removed from the model objective function by setting the **PV_carbon_cost** parameter to false on the **objective_function** tab.

Next, the user can update the **emissions_limit** tab to constrain emissions to the desired level and set the **emissions_limit** field on the **constraints_to_use** tab to TRUE.

Scenario E: Limit H2 and CCS to clusters

The user may be interested in simulating a scenario where Hydrogen and CCS deployment is limited to sites in Industrial clusters, rather than to all UK Industrial sites.

The cluster radius parameters **cluster_radius_H2** and **cluster_radius_CCS** in the **cluster_radius** tab allow the user to limit the distance from the cluster centres that H2 and CCS are available to sites (i.e. sites that fall outside this radius cannot access H2/CCS). A radius for each cluster centre of 999 km can be used to enable all sites to access H2 and CCUS.

Scenario F: Limiting the share of biomass for industry

The **max_fuel_constraints** tab allows the user to constraint the maximum amount of a particular fuel or fuel category that can be used in the model. For example, the user may be interested in exploring the impact of constraining the amount of biomass used in the model. Valid fuel names and fuel category names to use can be found on the "commodities" sheet.

A default *Biomass and organic waste* constraint is already included in the Input workbook and provides an example of constraining fuel using a **commodity_category** (i.e. group of fuel types such as heat, steam, coal; see column E on the **commodities** tab of the Input workbook).

Scenario G: Adding new technologies

COMIT makes it easy to amend or add to the list of technologies included in the model.

This can be done using the **Technologies** and **technology_input_output** tabs. In the **Technologies** tab, a new row can be added for each new technology to be incorporated into the model, that defines all the relevant information for that technology to be used in the model (e.g. capex, outputs, sector, lifetime).

For each new technology added, the corresponding additional input/output rows must be added to the **technology_input_output** tab.

An alternative method for turning technologies 'on' or 'off' in the model is to use the **start_year** parameter in the Technologies tab. Setting it to '2060' or a value larger than the model **end_year** (e.g. 2050) will mean that the Technology will never be deployed.

Scenario H: Limiting the rate at which industry sectors decarbonise

The **supply_chain_constraints** tab provides a way to constrain the amount of new capacity of technologies added in each year in each sector. It allows the user to limit the maximum rate at which each sector can decarbonise in each year and explore scenarios in which the rollout of emission reducing technologies is constrained by supply chain capacity (as would be expected to be the case in practice).

To be active, the corresponding '**constraints_to_include**' value must be set to TRUE, allowing the user to easily switch on or off the supply chain constraint.

Scenario I: Exploring the impact of Hydrogen network assumptions

The **H2_availability** tab can be used to set the maximum amount of Hydrogen available to each industrial cluster in each year. This allows the user to constrain the deployment of hydrogen at the cluster level.

This tab is designed to input the total Hydrogen production in the cluster minus the non-industrial Hydrogen, so users need to make sure they input H2 availability values *after* subtracting the amount of non-industrial hydrogen (see the **Non_industry_H2_demand** tab). To turn off this constraint, the user can simply set the **H2_availability** parameter in the **constraints_to_include** tab to FALSE.

Scenario J: Exploring impact of different CO2 networks and CO2 T&S costs.

The **CO2_transport_cost** sheet allows the user to set the available options and costs for transporting CO2 via trucking and pipelines. This is used by the model to calculate CO2 transportation costs between industrial sites. This tab allows the user to explore how CO2 trucking and pipeline options impact the cost and speed of decarbonisation.

The **lowest_cost_option** variable provides functionality that allows the user to pre-specify the preferred transport option for CO2 for each flow rate and distance combination (Trucking = 1, Pipeline = 2). For example, setting the **lowest_cost_option** to 2 for all rows would mean that only pipelines can be used to transport CO2 from sites to clusters, which may mean that CO2 transport becomes more expensive.

A similar worksheet, **H2_transport_cost** is available that allows the user to modify H2 transport costs and options in a similar fashion to the **CO2_transport_cost** sheet.

Scenario K: Implementing an optimism bias

Two **optimism bias** parameters are included to allow the user to account for the potential for undershoot in the early estimates of the capital technology and pipe costs (e.g. due to costs which might have been excluded from the scopes of cost estimates or underestimated)

The **tech_optimism_adjustment** parameter allows the user to specify the amount to multiply all technology capital expenditure costs by when accounting for this optimism bias. Similarly, the **pipe_optimism_adjustment** parameter allows the user to specify the amount to multiply all CO2 and H2 pipeline capital expenditure costs by.

Scenario L: Site-level access to the grid

The **H2_grid_start** tab allows the user to specify the first year at which a site can access hydrogen via connection to the national grid for each NAEI site included in the model. This information is specified by NAEI plant ID, and the model then maps these plant IDs on to the corresponding NAEI site IDs in the model.

An additional option has been added to allow the user to define the type of grid access scenario they wish to model for non-point source sites (non-point source sites are those that are not covered in the NAEI dataset). On the **model_parameters_b** tab set the parameter **use_nps_for_H2_grid_sites** to TRUE to allow non-point source sites to be able to access the grid, or FALSE to allow only point source sites to be able to access the grid. If **use_nps_for_H2_grid_sites** is set to TRUE, the model uses the average of the years that point source sites can access the grid (rounded up) in the same cluster and inside/outside cluster location as the non-point source sites.

Scenario M: Constrain electricity by cluster

The model allows the user to constrain the amount of electrical capacity available to sites (electricity 'headroom') by cluster and year. The model provides the user with the ability to switch between different headroom constraint options by using the **headroom_scenario** parameter on the **model_parameters_c** tab. This functionality includes the ability to select 'no constraint' (no limits on electrical capacity) and to specify a custom scenario by inputting the desired figures for the model to use into the relevant column on the **headroom_constraints** tab.

Scenario N: Limit switching between technologies

The **tech_stability** constraint allows the user to constrain the degree to which the model can 'switch away' from technologies currently in use. This constraint allows the user to ensure the model doesn't switch back and forth between different industrial technologies at each site (e.g. from gas technologies to electricity and then back), which would not be realistic in practice.

Users can add the technology code to Column A for each technology they want to limit switching behaviour for and then define the level of 'stability' required. The **stability_length** parameter defines the years of consistent use required, or how long to constrain the use of the technology for, and the **stability_factor** defines the proportion of new capacity that has to be used for the **stability_length** specified. A **stability_length** of 10 and **stability_factor** of 0.8, would mean that at least 80% of a technologies capacity needs to be used for the next 10 years.

Scenario O: Limiting the proportion of a fuel type that is used in each year

The **max_fuel_share** tab allows the user to constrain the proportion of a fuel type that is used in each year after the start year for each output in each sector. The input values for each fuel type are the amount of the fuel share for that output commodity that cannot be exceeded after the first year of the model.

This tab allows the user to constrain the model to ensure that fuel use is reflective of what might be expected in practice based on current usage trends and existing industrial technological capacity.

6. Making changes to the COMIT code

For users interested in viewing or making changes to the code underlying COMIT, this can be done using a copy of the latest COMIT code, which is available on the project Git repository. Once loaded in R, the app can be launched by running these two commands:

```
devtools::load_all()
```

```
run_app()
```

The **devtools::load_all()** command reloads all functions and libraries into the R environment. This command roughly simulates what happens when a package is installed and loaded with the library call (the usual way to load and run COMIT).

Appendix A

Interpreting technology codes

In COMIT, each industrial sector has been allocated a 3-letter code. For example, ICM is the code for the cement sector. In most cases, this also corresponds to the code for the final output commodity produced by that sector to meet industrial demand (the “demand commodity”). The exceptions to this are subsectors such as high value chemicals and printing, which have their own demand commodities. The demand commodities used in COMIT can be found in the **Demand_drivers** tab in the input workbook.

For technologies, the first 3 digits of the technology codes indicate which sector it is for. For many of the technologies, the next three digits then indicate the industrial process involved, and the three digits after that indicate the type of fuel being used. For example, IOIDRY indicates technologies for drying in the Other industries sector (IOI), and IOIDRYELC01 uses electricity (ELC) while IOIDRYNGA01 uses natural gas (NGA).

Table 2 List of the technology/process codes in COMIT.

Sector / subsector	Code	Process	Process codes	Other codes	
<i>Ammonia*</i>	IAM	High temperature heat	HTH	HP	Heat pump
<i>Chemicals</i>	ICH	Low temperature heat	LTH	SC	Steam cracker
<i>Cement</i>	ICM	Drying	DRY	KLM	Kiln
<i>Food & Drink</i>	IFD	Motoring	MOT	BIO	Boiler
<i>High value chemicals*</i>	IHVC	Refrigeration	REF	FUR	Furnace CHP
<i>Iron & Steel</i>	IIS	Other	OTH	CHP	technologies
<i>Non-ferrous metals</i>	INF	Low temp space heating	SPC	COG	Coke oven gas
<i>Ceramics</i>	ICR				
<i>Other</i>	IOI				
<i>Paper production</i>	IPP	Fuel	Fuel codes		
<i>Refineries</i>	PRCOIL	Natural gas	NGA		
<i>Glass</i>	IGL	Coal	HCO		
<i>Mechanical engineering</i>	IME	Biomass	BIO		
<i>Electrical engineering</i>	IEE	Liquified petroleum gas	LPG		
<i>Vehicles</i>	IVH	Hydrogen	HDG		
<i>Textiles</i>	ITX	Electricity	ELC		
<i>Construction</i>	ICN	Low-temperature steam/heat from CHP	STM		
<i>Printing*</i>	IPR				
<i>Lime</i>	ILM				
<i>Hydrogen</i>	PHYG				

**Indicates a subsector*